# RAHUL TANGSALI

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## Education

**University of Southern California - M.S. in Computer Science (08/2023 - 05/2025)** Los Angeles, CA
**Pune Institute of Computer Technology - B.E. in Computer Engineering (08/2019 - 06/2023)** Pune, India

## Technical Skills

**Languages:** Python, C++, TypeScript, JavaScript, SQL
**AI/ML:** PyTorch, Hugging Face Transformers, LLMs, RAG, LangGraph, LangChain, NLP, Deep Learning, BERT, mBERT, Embeddings, Fine-Tuning, Model Evaluation
**AI Infrastructure:** Pinecone, Cohere, OpenAI, MCP, Vector Search, Reranking, LLM-as-Judge
**Engineering:** FastAPI, Node.js, Express.js, Next.js, React, Angular, MongoDB, PostgreSQL, MySQL, Docker, AWS, GCP, Git

## AI/ML Projects

**MedAssist AI – Agentic Medical Assistant** *LangGraph, FastAPI, Next.js, Pinecone, OpenAI*

- Built an **agentic AI system** using **LangGraph and FastAPI** with **8 dynamically routed tools** integrating NIH, FDA, and NPI APIs, achieving **94.3% tool-routing accuracy** across 35 evaluation cases.
- Engineered a cascading retrieval pipeline combining **domain APIs, Pinecone RAG over 23,489 chunks, and LLM fallback**, with source-aware responses and real-time SSE streaming.
- Built an **LLM-as-Judge evaluation framework** across 11 categories, achieving **4.37/5 correctness**; shipped JWT authentication, user profiles, and a production-style Next.js interface.

**LLM Research Assistant – RAG-Powered Document Intelligence** *LangGraph, LangChain, OpenAI, Pinecone, Cohere, Next.js*

- Built a **full-stack RAG system** for conversational document analysis, achieving **93.75% faithfulness** and **97.32% answer correctness** across a 20-case LLM-as-Judge evaluation suite.
- Engineered a multi-stage retrieval pipeline with **OpenAI embeddings, Pinecone vector search, Cohere reranking, LLM-based document grading, and web-search fallback**, with MCP tool integrations and streaming responses.

**Continual Learning for Multilingual Named Entity Recognition** *Python, PyTorch, mBERT, Hugging Face, WikiANN*

- Investigated **catastrophic forgetting in multilingual NER** by sequentially fine-tuning **mBERT** across English, Spanish, and Hebrew using the WikiANN dataset and comparing continual-learning strategies.
- Applied **Experience Replay and learning-rate adjustment** to preserve previously learned language representations; Experience Replay achieved F1 scores of **0.811 English, 0.911 Spanish, and 0.824 Hebrew**.
- Evaluated cross-lingual knowledge retention and transfer using **F1, Backward Transfer (BWT), and Forward Transfer (FWT)**, reducing English forgetting to approximately **0.019 F1** under Experience Replay.

## Experience

**OneHermes** Madison, WI
Founding Software Engineer 06/2026 – Present

- Shipped **300+ tickets** regarding new features, production enhancements and defect fixes for an enterprise workforce and healthcare platform using **Angular, Node.js, Express, MongoDB, and REST APIs**.
- Developed and tested **EVV integrations and Patient Care workflows**, implementing API-driven healthcare data processing, validation, and end-to-end frontend/backend functionality.
- Worked directly with stakeholders across **3+ customer environments**, owning requirements and reported issues from **root-cause analysis through implementation, testing, release validation, and client follow-up**.

**University of Southern California** Los Angeles, CA
Research Software Engineer 08/2025 – 06/2026

- Built **Tommy**, an AI-powered virtual assistant deployed on the USC OIS website to answer F-1/J-1 immigration questions **24/7**, handling **2,000+ student queries/month**.
- Developed automated ingestion and web-scraping pipelines across **15+ university and government sources** to maintain an up-to-date knowledge base for AI-generated responses.
- Designed a **human-in-the-loop evaluation pipeline** using user feedback and logged conversations to identify failure cases and iteratively improve chatbot response quality.

**L3Cube** Pune, India
Machine Learning Research Intern 06/2022 – 06/2023

- Built and evaluated a **60K+ sample multi-domain Marathi sentiment corpus** spanning movie reviews, social media, subtitles, and political text, with human annotation achieving **0.86 Cohen's Kappa**.
- Fine-tuned and benchmarked **MahaBERT, mBERT, MuRIL, and IndicBERT** for low-resource sentiment classification, with Marathi-specific MahaBERT achieving up to **84.9% accuracy**.
- Researched **data augmentation and cross-domain generalization** using back-translation, paraphrasing, BERT-based masking, NER masking, and GPT-based generation to improve robustness for low-resource NLP.